

42.510 Statistical Learning for Data Science

Classification - beyond logistic regression

Bikramjit Das

Week 4, Lecture 1

Challenger disaster

- Predict O-ring failure using Temperature and Pressure.
- Used a Logistic regression model.
- Estimated parameters β .
- **Quality of fit:**
 - ▶ **Null deviance:** fit logistic regression with only the intercept term.
Estimate β_0 such that $\mathbb{P}(\widehat{Y} = 1) = \bar{y} = \frac{e^{\beta_0}}{1 + e^{\beta_0}}$.
 - ▶ Then Null deviance $= -2LL(\hat{\beta}_0)$.
 - ▶ **Residual deviance** from full model $= -2LL(\hat{\beta})$.
 - ▶ Clearly residual deviance is lower than null deviance and keeps getting lower as we increase the number of dimensions p .
- **Akaike's information criterion (AIC):**

$$\text{AIC} = -2LL(\hat{\beta}) + 2(p + 1).$$

Quality of fit: confusion matrix

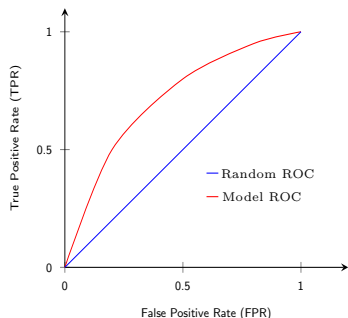
- Predict based on a threshold t and the estimated $\hat{\beta}$.

$$\begin{aligned}\mathbb{P}(Y_i = 1 | \mathbf{x}_i, \hat{\beta}) \geq t, & \text{ predict } \hat{Y}_i = 1, \\ < t, & \text{ predict } \hat{Y}_i = 0.\end{aligned}$$

	Actual = 0	Actual = 1
Predict = 0	True Negative (TN)	False Negative (FN)
Predict = 1	False Positive (FP)	True Positive (TP)

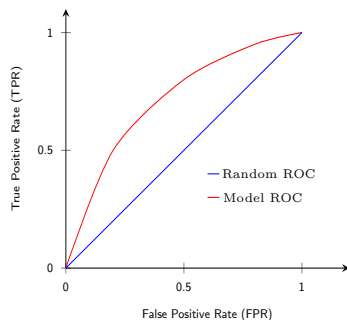
- True negative rate: $\text{TNR} = \frac{TN}{FP+TN}$ (Specificity),
- False positive rate: $\text{FPR} = \frac{FP}{FP+TN}$ (Type I error),
- True positive rate: $\text{TPR} = \frac{TP}{FN+TP}$ (Sensitivity),
- False negative rate: $\text{FNR} = \frac{FN}{FN+TP}$ (Type II error),
- Overall Accuracy: $\text{Accuracy} = \frac{TN+TP}{FP+TN+FN+TP}$.

Quality of fit: Receiver operating characteristic (ROC) curve



- Previously we fixed t and found the **confusion matrix**.
- **ROC**: Plot $(FPR(t), TPR(t))$ for $0 \leq t \leq 1$.
- $t = 0$: By definition, $TN = FN = 0$ and hence $FPR = TPR = 1$.
- $t = 1$: By definition, $FPR = TPR = 0$.
- Often a **baseline** model uses $t = 0.5$.

Quality of fit: Area under the curve (AUC)



- Area under the curve: **AUC**.
- **AUC** for the random ROC = 0.5.

$\text{AUC} = \mathbb{P}(\text{a randomly chosen point with } Y = 1 \text{ is ranked higher than a randomly chosen point with } Y = 0).$

Classification

- Logistic regression
- Probit regression - replace logit by normal
- Linear Discriminant Analysis (LDA)
- Quadratic Discriminant Analysis (QDA)
- Perceptron
- K-NN
- Decision trees

Linear Discriminant Analysis (LDA)

- In **Logistic Regression** we modeled $\mathbb{P}(Y = y|\mathbf{X} = \mathbf{x})$.
 - ▶ For two classes with $y \in \mathcal{Y} = \{0, 1\}$ we had

$$\mathbb{P}(Y = 1|\mathbf{x}, \beta) =: p(\mathbf{x}, \beta) = \frac{\exp(\beta^\top \mathbf{x})}{1 + \exp(\beta^\top \mathbf{x})},$$

$$\mathbb{P}(Y = 0|\mathbf{x}, \beta) = 1 - p(\mathbf{x}, \beta) = \frac{1}{1 + \exp(\beta^\top \mathbf{x})}.$$

- In **LDA**
 - ▶ we model $\mathbb{P}(\mathbf{X} = \mathbf{x}|Y = y)$ for each class $y \in \mathcal{Y} = \{1, 2, \dots, K\}$.
 - ▶ let π_k be prior probability of each class $y \in \mathcal{Y}$.
 - ▶ use Bayes' theorem to find $\mathbb{P}(Y = y|\mathbf{X} = \mathbf{x})$.

Linear Discriminant Analysis (LDA)

- Using Bayes' theorem we have for $k \in \mathcal{Y} := \{1, \dots, K\}$,

$$p_k(\mathbf{x}) = \mathbb{P}(Y = k | \mathbf{X} = \mathbf{x}) = \frac{\pi_k f_k(\mathbf{x})}{\sum_{\ell} \pi_{\ell} f_{\ell}(\mathbf{x})}.$$

- π_k is estimated from the data if we had a random sample from the population.
- For estimating f_k we impose some model assumptions.
- Idea:** Estimate f_k such that we approximate the Bayes classifier with 0-1 loss, which has the lowest error rate out of all classifiers.

$$\hat{Y}(\mathbf{x}) = \arg \min_{k \in \mathcal{Y}} \sum_{\ell=1}^K L(\ell, k) \mathbb{P}(Y = \ell | \mathbf{X} = \mathbf{x}).$$

- This implies we classify to the class for which $p_k(\mathbf{x})$ is the largest:

$$\hat{Y}(\mathbf{x}) = \arg \max_{k \in \mathcal{Y}} \mathbb{P}(Y = k | \mathbf{X} = \mathbf{x}).$$

LDA with $p = 1$

- Assume that f_k is normal/Gaussian, hence its **pdf** is

$$f_k(x) = \frac{1}{\sqrt{2\pi}\sigma_k} e^{-\frac{1}{2\sigma_k^2}(x-\mu_k)^2}$$

where μ_k and σ_k^2 are the unknown mean and variance for the k th class.

- Assume

$$\sigma_1^2 = \dots = \sigma_k^2 = \sigma^2 \text{ (say).}$$

- Therefore

$$p_k(x) = \frac{\pi_k e^{-\frac{1}{2\sigma^2}(x-\mu_k)^2}}{\sum_{\ell=1}^K \pi_\ell e^{-\frac{1}{2\sigma^2}(x-\mu_\ell)^2}}.$$

For any $X = x$, *Bayes classifier* chooses k with largest $p_k(x)$.

- Thus (after taking logarithms), $\hat{Y}(x) = \arg \max_{k \in \mathcal{Y}} \delta_k(x)$ where

$$\delta_k(x) = x \frac{\mu_k}{\sigma^2} - \frac{\mu_k^2}{2\sigma^2} + \log(\pi_k).$$

- $\delta_k(x)$'s are **discriminant functions**.
- $\delta_k(x)$'s are **linear functions** of x . Hence we call this LDA.

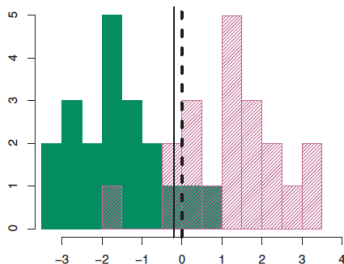
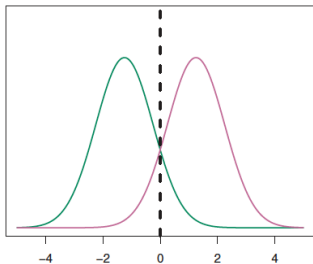
LDA with $p = 1$

- If $K = 2$ and $\pi_1 = \pi_2$, then the *Bayes decision boundary* is at:

$$x^* = \frac{\mu_1 + \mu_2}{2},$$

and we have

$$\hat{Y}(x) = \begin{cases} 1, & \text{if } 2(\mu_1 - \mu_2)x > \mu_1^2 - \mu_2^2, \\ 2, & \text{otherwise.} \end{cases}$$



LDA with $p = 1$

- In practice, we do not know π_k, μ_k, σ^2 and need to estimate them.
- We have data: $(x_1, y_1), \dots, (x_n, y_n)$ where $y_i \in \mathcal{Y}, x_i \in \mathbb{R}$ for $i = 1, \dots, n$.
- We use the following plug-in estimates:

$$\hat{\pi}_k = \frac{n_k}{n} \quad \text{where } n_k = \text{number of data points from class } k,$$

$$\hat{\mu}_k = \frac{1}{n_k} \sum_{i:y_i=k} x_i,$$

$$\hat{\sigma}^2 = \frac{1}{n - K} \sum_{k=1}^K \sum_{i:y_i=k} (x_i - \hat{\mu}_k)^2.$$

- Therefore we have

$$\hat{Y}(x) = \arg \max_{k \in \mathcal{Y}} \hat{\delta}_k(x)$$

where

$$\hat{\delta}_k(x) = x \frac{\hat{\mu}_k}{\hat{\sigma}^2} - \frac{\hat{\mu}_k}{2\hat{\sigma}^2} + \log(\hat{\pi}_k).$$

LDA with $p > 1$

- We have p predictor variables instead of 1: $\mathbf{X} = (X_1, \dots, X_p)$.
- Assume that data $\mathbf{X}$ for each class $k \in \mathcal{Y}$ is generated from a multivariate normal/Gaussian distribution f_k with possibly unequal mean vector $\boldsymbol{\mu}_k$ but same Covariance matrix Σ :

$$f_k(\mathbf{x}) = \frac{1}{(2\pi)^{p/2}|\Sigma|^{1/2}} \exp\left(-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu}_k)^\top \Sigma^{-1}(\mathbf{x} - \boldsymbol{\mu}_k)\right).$$

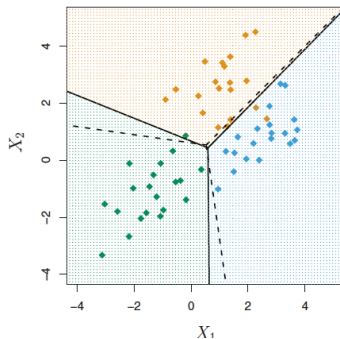
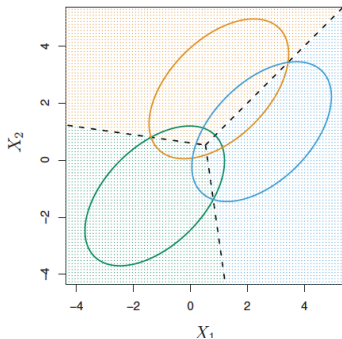
- Bayes classifier assigns observation $X = x$ to the class whose discriminant function $\delta_k(\mathbf{x})$ is largest:

$$\delta_k(\mathbf{x}) = \mathbf{x}^\top \Sigma^{-1} \boldsymbol{\mu}_k - \frac{1}{2} \boldsymbol{\mu}_k^\top \Sigma^{-1} \boldsymbol{\mu}_k + \log \pi_k$$

which is again a linear function of $\mathbf{x}$.

LDA with $p > 1$

- We need to estimate all the parameters μ_k, Σ, π_k from the data (in a similar manner, e.g., using plug-in estimates).

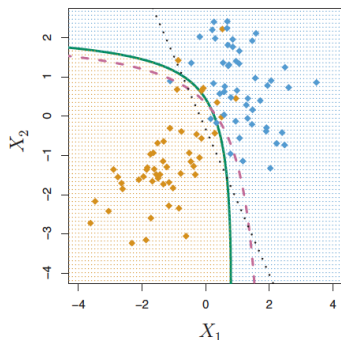
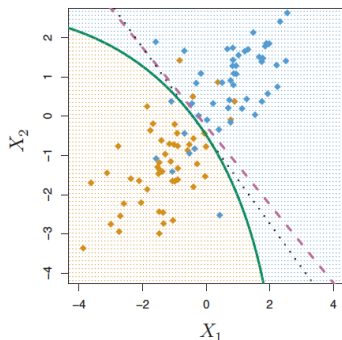


Quadratic Discriminant Analysis (QDA)

- Difference from LDA: we allow the covariance matrices to be unequal.
- The discriminant function becomes:

$$\delta_k(\mathbf{x}) = -\frac{1}{2}\mathbf{x}^\top \Sigma_k^{-1}\mathbf{x} + \mathbf{x}^\top \Sigma_k^{-1}\boldsymbol{\mu}_k - \frac{1}{2}\boldsymbol{\mu}_k^\top \Sigma_k^{-1}\boldsymbol{\mu}_k - \frac{1}{2}\log |\Sigma_k| + \log \pi_k$$

which is now a **quadratic function** of $\mathbf{x}$. Hence the name QDA.



Comparison: Logistic vs LDA vs QDA

	Aspect	Logistic	LDA	QDA
1	Decision boundary	Linear	Linear	Quadratic
2	Covariance	Not relevant	Same for all classes	Different for each class
3	Addressing non-linearity	Cannot handle directly	Assumes linear separability	Through quadratic boundaries
4	Computational complexity	Moderate (solve via Newton Rhapson)	Moderate (estimating covariance)	Higher (estimating covariance for each class)
5	Ideal case	Data is linearly separable	Data is linearly separable with homogeneous covariances	Data has complex, non-linear class boundaries and different class variances

Binary classification

- Create confusion matrix.

	Actual = 0	Actual = 1
Predict = 0	True Negative (TN)	False Negative (FN)
Predict = 1	False Positive (FP)	True Positive (TP)

- All methods use a default threshold of 0.5.
 - ▶ You may want to change this according to your use.
- Plot ROC curve and compute AUC (a measure of model fit).

What should we use?

- Depends on the problem; we often try all (including KNN and others).
- Balanced problem, no particular distributional assumption: try logistic.
- Imbalanced problem, some idea about generative distribution: try LDA/QDA.

References

- Hastie et al. (2001), *The Elements of Statistical Learning*, Chapter 4.
- James et al. (2023), *Introduction to Statistical Learning*, Chapter 4.